

MATH 403, 1st Examination Review Problems

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1. Show that for any odd integer n , $n^2 = 1$ modulo 8. What does this say about the structure of the group $(\mathbb{Z}/8)^\times$ (or $U(8)$ in Gallian's notation)?
2. Show that the groups $GL(2, \mathbb{Z}/2)$ and S_3 are isomorphic.
3. (Gallian, Ch. 3, #92) Let G be a finite group with more than one element. Show that G has an element of prime order.
4. (Gallian, Ch. 5, #55) In S_4 , find a cyclic subgroup of order 4 and a noncyclic subgroup of order 4.
5. If G is an abelian group of odd order, show that $x \mapsto x^2$ is an automorphism of G . Where does the odd order assumption get used?
6. Give an example of a nonabelian group of order 8.
7. Show that there is no nonabelian group of order 33.